

K0maru-Agent-Memory: 4-Layer Decoupled Architecture

High-performance, single-static-binary, zero-daemon memory architecture for autonomous coding agents

Layer 4: Agent Protocol & Ecosystem Integration

k0maru::cli · k0maru::mcp · k0maru::ecosystem

FastMCP Stdio Server

- Official MCP JSON-RPC 2.0 protocol
- Zero open ports · pure stdio pipes
- Adaptive intent-driven tool routing

CLI Command Suite

- ``k0maru loadout`` (<300 token budget)
- ``k0maru offload`` & ``k0maru inspect``
- ``k0maru search``, ``flush``, ``distill``

Ecosystem Connectors

- One-click auto-configurator & doctor
- Claude Code · Cursor · Windsurf
- Cline · Hermes · OpenClaw

Embedded Web Cockpit

- ``k0maru ui`` on local loopback
- Developer Dark Cockpit theme
- Search debugger & token scoreboard

Layer 3: Hybrid Retrieval & Explainable Ranking Engine

k0maru::storage::hybrid · k0maru::vector

SQLite FTS5 BM25

- BM25 lexical ranking for exact code tokens
- Language-tolerant prefix & symbol tokenization
- Sub-millisecond query execution

sqlite-vec Dense Vectors

- Local 384-dim vector virtual table
- Normalized cosine distance search
- Pure C/Rust embedded, zero external services

FastEmbed ONNX Inference

- Local CPU ``all-MiniLM-L6-v2`` model
- Quantized ~23MB footprint, zero GPU req
- Deterministic offline vector generation

RRF Fusion + Graph Boost

- Reciprocal Rank Fusion (k=60)
- +0.05 1-hop bidirectional boost
- Elevates contextually linked notes

Layer 2: Incremental File Scanner & Transient Cache

k0maru::scanner · k0maru::convention · k0maru::offload

Incremental Scanner

- Dual-key mtime + xxh3 dirty check
- Zero re-computation on untouched notes
- <12ms full vault pass (<75ms for 500 docs)

Convention Sniffer

- Adaptive vault topology sniffer
- Heuristic folder & naming style detector
- Frontmatter & template auto-synthesizer

Disposable Cache

- Transient ``k0maru/cache.sqlite``
- Strictly disposable & non-authoritative
- 74ms instant wipe & rebuild

AST Offload Engine

- Streaming shell pipe threshold filter
- Error signature regex & AST extractors
- Deterministic Mermaid state chart gen

Layer 1: Knowledge Storage & Markdown Adapters

k0maru::adapters · k0maru::parser

Obsidian Vault Adapter

- Full folder taxonomy support
- YAML frontmatter overrides
- Strict hierarchy tag extraction

Generic LLM-Wiki Adapter

- Karpathy flat wiki structure
- Custom subdirectories & roots
- Zero proprietary file formats

pulldown-cmark Parser

- AST Markdown streaming parser
- Codeblock link suppression
- Bare & aliased ``[[WikiLinks]]``

Plain-Text Markdown Vault

- Human & Git-native ownership
- Local file system as single source of truth
- Permanent knowledge compounding